Fiscal Year:	FY 2013	**Task Last Updated:**	FY 05/07/2013
PI Name:	Kamotani, Yasuhiro Ph.D.		
Project Title:	Investigation into the Quantification and Control of Steady and Unsteady Flow Phenomena in Passive Gas-Liquid Cyclonic Separators		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Sharp, Lauren	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date is now 8/31/2014, per NSSC information (Ed., 6/10/14) NOTE: End date is now 5/31/2014, per J. McQuillen/GRC (Ed., 3/7/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kadambi, Jaikrishnan (Case Western Reserve University)		
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Task Description:

Phase separators are devices designed to divide flows of intermingling gas and liquid into separate streams that are individually purely liquid or purely gas. Separation of the respective phases is of considerable interest to NASA as many spacecraft systems can only function effectively if the flow they receive is composed exclusively of the one phase that they are designed to accommodate. While separation occurs naturally on the earth, the absence of buoyancy in a microgravity environment allows the undesired mixtures of gas and liquid to remain combined unopposed. Due to this reluctance of the phases to separate, installation of some type of separation device becomes supremely important for many of the fluid systems that are used by NASA. Cyclonic separators provide the gas-liquid separatory action by swirling the multiphase flow – causing the gas to accumulate along the axis of the vortex as the denser liquid is forced to the walls – thereby allowing segregated extraction of the respective phases. Passive cyclonic separators utilize only the inertia of the incoming flow to accomplish this task and in doing so can avoid the moving parts and reliability concerns of their active brethren. While devices of this type have been successfully employed in experimental settings, more complete investigations and analyses are required before these devices can ever be seriously considered for use in operational space hardware.
The objective of this study is to facilitate the maturation of passive cyclonic separation technology through exhaustive research of a newly designed gas-liquid phase separator evolved from NASA Glenn Research Center's (GRC) existing Cascade Cyclonic Separation Device (CSD-C). Combined experimental and numerical analyses will seek to quantitatively assess and delimit the range of operability of the new separator including the transient and steady-state instability behavior of the device with respect to both device and system-level conduct – features that have yet to be completely explored in any such device.

The approach is first to assess the range of operability of the new separator design through the careful construction of detailed flow separation maps using data gathered from reduced-gravity flight experimentation in conjunction with ancillary computational studies. Second, the transient and stability behavior of the device will be observed and analyzed. Special attention will be paid both to device-level conduct and also to behaviors that could manifest dysfunction elsewhere at the complete system level. A multiscale hybrid of the volume-of-fluid (VOF) and Euler methods has been coupled to 1-D system-level multiphase solvers in order to permit discernment of these flow features. Lastly, with the ensuing understanding of the causes of any performance-affecting responses, new ways to control transient and unstable operations will be sought; in particular, the insight acquired should facilitate further new and original design modifications to improve the performance of microgravity phase separation devices. Validation and verification of the numerical results are expected to permit the use of CMFD as a reliable design tool to further assist in the creation and analysis of new designs.

The combination of the research activity included in this grant with the complementary activity of Dynaflow Inc. of Jessup, MD (Grant NNX09AI35G) has evolved into the Two-Phase Flow Separator Experiment (TPFSE). TPFSE represents a comprehensive investigation of passive cyclone separation technology – an investigation made possible by capabilities that only NASA and the International Space Station can offer.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Phase separation has been identified as a critical mission-enabling capability for spacecraft systems ranging from power conversion to Environmental Control and Life Support Systems. As these systems have little tolerance for failure, extensive investigations and analyses are required before these devices can ever be seriously considered for employment in operational space hardware. Due to the advantages associated with passive vortex devices, cyclonic separators are also finding increasing use in terrestrial applications, such as in the petroleum industry where separators of this type are considered to be an emerging technology. The detailed study of the features of the complicated vortex-driven flow along with the modeling techniques under development for the microgravity investigations could equally be exploited for the advancement of these more Earthly applications.

Task Progress:

Experimentation
Experimentation with various vortex separation devices has been conducted throughout the past year of this grant in both terrestrial and reduced-gravity conditions.

The main work consisted of the testing of various separator designs during a reduced-gravity flight campaign. The separator was installed in the Two-Phase Flow Experiment apparatus from NASA Glenn Research Center. Various diagnostic techniques, including electrical resistance tomography (ERT) and high speed imaging, were used to classify the performance of the separators. To achieve reduced-gravity conditions the rig was installed aboard a Zero-G Corp. Boeing 727 and flown in parabolic arcs.

The reduced-gravity flights could be used to determine the performance of the separator at flow rates that are too slow to be tested in terrestrial conditions (due to the preponderance of gravity). In general, the steady performance was found to be similar to operations in terrestrial conditions. Various stages of unsteady performance could also be examined, including separator shutdown (which is a stage of the flow that cannot be examined in 1-g).

Numerical and Analytical Modeling

The CMFD studies of the fluid physics within the separation device conducted during the previous years of the grant have been continued. A hybrid VOF-multifluid approach has been created allowing for the simulation of both the small-scale separation of the injected bubbles and the large-scale capillary forces at the gas core.

The control-volume-based analytical approach (initially developed in previous years) has been extended to account for multiphase injection into the separator. The modifications required the addition of terms to account for backpressure at the gas outlet and a complete model of the nozzle to handle inhomogeneous injection. The analytical approach has continued to provide insight into device design. The model has been compared to results for both reduced-gravity and terrestrial experiments and the results have been found to be good.

TPFSE

This research activity has evolved into the Two-Phase Flow Separator Experiment (TPFSE). In accordance with the elevated status, the definition of the complete experimental requirements has been continued with respect to the objective of the emplacement of the experimental apparatus aboard the International Space Station. On May 15, 2013, the project Science Concept Review is scheduled to occur. After that, collaboration with NASA engineers will begin

	regarding the engineering design of the space experiment. Preparation for the eventual flight experiment will be continued in the coming year in concert with the above studies.
Bibliography Type:	Description: (Last Updated: 03/06/2024)
Abstracts for Journals and Proceedings	Hoyt N, Kadambi J, Kamotani Y. "The Performance of Passive Cyclonic Two-Phase Separators in Reduced Gravity." 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Program and abstracts, p. 54. , Nov-2012
Abstracts for Journals and Proceedings	Kang M, Lee K, Hoyt N, Kadambi J, Kamotani Y. "Steady and Dynamic Behaviors of Gas Core in a Passive Cyclonic Separator." 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Program and abstracts, p. 103. , Nov-2012
Abstracts for Journals and Proceedings	Kharraz A, Hoyt N, Kadambi J, Kamotani Y. "System Code for Two-Phase Flow in Microgravity." 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Program and abstracts, p. 104. , Nov-2012
Dissertations and Theses	Hoyt NC. "The Performance of Passive Cylconic Separators in Microgravity." Dissertation, Case Western Reserve University, Cleveland, Ohio, August 2013. , Aug-2013